Buoyancy: Weighing Water With a Piece of Metal

SYNOPSIS

Hang a metal cylinder from a spring scale and read its weight. Now lower it into water, without letting it touch the bottom, and read the scale again. It reads less. Nothing has been removed from the cylinder, so the missing weight has to be a force — the buoyant force — pushing up on it. Archimedes' principle says that force equals the weight of the water the cylinder pushed out of the way, and this lab turns that statement into a measurement. The elegant part is what happens when you put the two halves together algebraically: the volume of displaced water appears on both sides and **cancels**, leaving the density of water expressed in terms of nothing but the density of the metal and the two scale readings. You never need to measure the water at all.

Due see the Module 6 Lab folder on Brightspace | **Report** Full lab report — use the Lab Template | **Pairs with** M6 N1 | **Data** provided — this is a virtual lab

PERFORMANCE EXPECTATIONS what you should be able to do when this lab is written up

- **PE1** **Draw the free-body diagram for a submerged object hanging from a scale** — weight down, tension up, buoyant force up — and write the equilibrium condition it expresses.
- **PE2** **Derive $F_B = T_{air} - T_{water}$** from that diagram, and explain why the scale reading is a tension rather than a weight.
- **PE3** **Obtain a volume by displacement** from two graduated-cylinder readings, and state the precision that method actually carries.
- **PE4** **Show algebraically that the displaced volume cancels**, arriving at $\rho_{water} = \rho_{metal}(1 - T_{water}/T_{air})$.
- **PE5** **Compute a density two different ways and compare them**, recognising when two routes share a common measurement and are therefore not independent.
- **PE6** **Trace a discrepancy to the measurement that caused it**, using the identity of the metal as an external check on your own data.

Start from Note 1 — this lab is that note, measured

M6 N1 is not background reading for this lab. It is the lab. Pressure in a fluid, the origin of the buoyant force in the pressure difference across an object's top and bottom faces, and Archimedes' principle itself are all established there. What the note cannot give you is a set of real scale readings that do not quite produce 1.000 g/cm^3. That is what you have here, and the gap between "water has a density of 1" and "my measurement says 0.91" is what this report is about.

Where each part of Note 1 belongs in this report. The rubric awards points for these.

Report section	What Note 1 gives you

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Introduction	Archimedes' principle stated in your own words and $F_B = \rho_{fluid} V g$ with every symbol defined, plus **your own derivation** of the working relation in which the volume cancels. End with a hypothesis you can confirm or reject.
Experimental	The labelled sketch of the cylinder on the scale, in air and in water, with the forces marked in both cases. Say how the displaced volume was read and to what precision.
Analysis	The buoyant force from the two scale readings, the density of the metal from mass and displaced volume, and the density of water by both routes.
Discussion	Why the two routes disagree, which measurement is responsible, and what the identity of the metal tells you that your own numbers cannot.

Read or reprint the note here: Module 6 Note 1 — Fluids, Pressure and Buoyancy.

The physics you are using

In air, and then in water

Hanging in air, the cylinder is in equilibrium under two forces, so the scale reads its weight:

$$T_{air} = F_g = M_{metal}\, g$$

Submerged, a third force appears. The cylinder is still in equilibrium, so now

$$F_B + T_{water} = F_g \quad \rightarrow \quad F_B = F_g - T_{water} = T_{air} - T_{water}$$

The buoyant force is the drop in the scale reading. Nothing else is needed to measure it.

The cancellation

Archimedes' principle says that same force is the weight of the displaced water:

$$F_B = \rho_{water}\, V_{displaced}\, g$$

Set the two expressions for F_B equal, and write $T_{air} = M_{metal} g$:

$$\rho_{water}\, V_{disp}\, g = T_{air} - T_{water} = T_{air}\left(1 - T_{water}/T_{air}\right) = M_{metal}\, g \left(1 - T_{water}/T_{air}\right)$$

The cylinder is fully submerged, so the water it pushed aside has exactly the cylinder's own volume: $V_{disp} = V_{metal}$. And $M_{metal} = \rho_{metal} V_{metal}$. Substitute, and **both the volume and g cancel from every term**:

$$\rho_{water} = \rho_{metal}\left(1 - T_{water} / T_{air}\right)$$

> **Look at what that equation does not contain.** No volume. No g. No unit conversion between newtons and grams. The density of water comes out of the density of the metal and a *ratio* of two scale readings — and because it is a ratio, any

constant calibration error in the scale cancels out of it too. This is the result the lab is built around, and deriving it yourself is worth more marks than any arithmetic below.

Predict first. You repeat the whole experiment with a cylinder of the same metal but twice the volume. What happens to the value of ρ_{water} you calculate?

(A) Unchanged — the volume cancels, and the ratio T_{water}/T_{air} is the same.

(B) It doubles, because twice the water is displaced.

(C) It halves.

(D) It cannot be predicted without knowing the new volume.

Why that question matters for how you write this up. Both scale readings double, so their ratio is untouched, and ρ_{metal} is a property of the material rather than of the lump. The predicted density of water is therefore identical. That is a strong statement: it says the experiment does not depend on getting the size of the cylinder right, only on the two scale readings and on knowing what the metal is. Keep that in mind when you reach the Discussion, because the number you actually compute will come out about 9% low — and if the method is insensitive to volume, the volume had better not be where you lay the blame without checking.

The apparatus

A metal cylinder from the lab kit hangs by a short length of string from a spring scale. Below it sits a graduated cylinder about half filled with water. The scale is read with the cylinder hanging freely in air, and again with it fully submerged and clear of the bottom and sides. The water level is read before and after each dip.

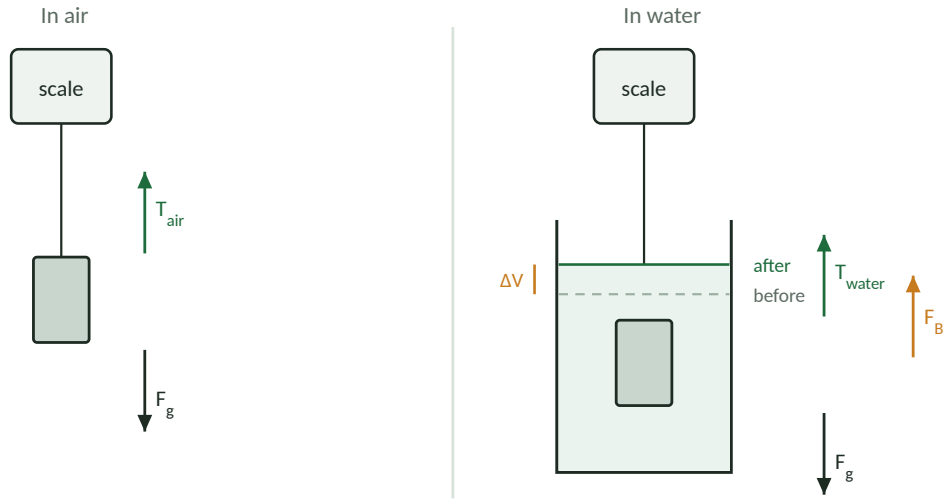


Figure 1. The two readings the whole lab rests on. In air the scale reads the full weight; submerged it reads less by exactly the buoyant force, and the water level rises by the cylinder's own volume.

Part I — The data

One cylinder, three dips. The mass was measured on a digital balance; the scale readings are in newtons.

The three trials as recorded, with the averages used in the analysis.

Trial	Water before (mL)	Water after (mL)	Weight in air (N)	Weight in water (N)

Trial	Water before (mL)	Water after (mL)	Weight in air (N)	Weight in water (N)
1	60.0	67.0	0.745	0.68
2	60.0	66.5	0.740	0.66
3	60.0	67.0	0.720	0.67
Average	**60.0**	**66.83**	**0.735**	**0.670**

The mass of the metal cylinder, from the digital balance, is **70.5 g**.

> **Notice the precision mismatch before you start calculating.** The balance gave the mass to a tenth of a gram — four significant figures. The graduated cylinder was read to the nearest half millilitre on a rise of under seven millilitres, which is two significant figures at best. Those two numbers are about to be divided by one another, and the weaker one governs the result. Carrying 70.5/6.83 to five digits does not make the answer more precise; it only hides which measurement is in charge.

Part II — The buoyant force and the metal

I. **The displaced volume.** Subtract the average before-reading from the average after-reading. 1 mL = 1 cm^3.

II. **The density of the metal.** $\rho_{metal} = M_{metal} / V_{displaced}$, since a fully submerged solid displaces its own volume.

III. **Identify the metal.** Compare your density against a table of common metals and name the best match. **This is not decoration** — it is the only external check in the whole experiment, and Part III turns on it.

IV. **The buoyant force.** $F_B = T_{air} - T_{water}$, in newtons.

Part III — The density of water, three ways

You now have enough to get ρ_{water} by more than one route, and they will not agree. Working out why is the point of this lab.

Route A — straight from Archimedes

$$\rho_{water} = F_B / (V_{displaced} \cdot g)$$

Watch the units: F_B in newtons and V in cubic metres gives kg/m^3; divide by 1000 for g/cm^3.

Route B — the cancellation, using your own metal density

$$\rho_{water} = \rho_{metal} (1 - T_{water}/T_{air})$$

Route C — the cancellation, using the published density of the metal you identified

Same formula as Route B, but with ρ_{metal} taken from a reference table for the metal you named in Part II rather than from your own displacement measurement.

> **Routes A and B are not independent, and that is the trap.** Both of them contain your displaced volume — A uses it directly, B hides it inside $\rho_{metal} = M/V$. So if the volume reading is off, both answers are off, and their rough agreement with each other proves nothing at all. **Route C is the only one that does not touch your volume measurement.** If C lands close to 1.000 g/cm^3 while A and B do not, you have located the faulty measurement without needing anyone to tell you. Your

> Discussion must address this explicitly; saying "the two methods agreed" when they share an input is a misreading of your own data.

What counts as agreement

The accepted density of water is **1.000 g/cm³**, or 1000 kg/m³. Report a percent error for each route. Within **5%** is a good result for this apparatus; outside **10%**, name the measurement responsible rather than calling it experimental error.

Check your analysis

Enter your averages to check the arithmetic of all three routes at once. Your report still has to show each calculation worked by hand with units.

TOOL · THE DENSITY OF WATER, THREE WAYS

Mass of the metal cylinder (g)

70.5

Displaced volume (cm³)

6.83

Weight in air T_{air} (N)

0.735

Weight in water T_{water} (N)

0.670

Published density of the metal you identified (g/cm³) — lead 11.34, copper 8.96, brass 8.5, steel 7.85, aluminium 2.70

11.34

Uses g = 9.8 m/s² in Route A. Routes A and B both depend on your displaced volume; Route C does not. That difference is the lab.

Additional Questions

Choose three of the five below and answer those three fully. Name which three you chose. Answer them in a section of their own headed **Additional Questions**, in **sentences and paragraphs** with the working shown, never as a bare numbered list and never by restating the question before answering it. Depth on three beats coverage of five; this section is worth 8 points on the rubric.

AQ1 — find the broken measurement. Route C uses the published density of the metal and avoids your volume entirely. Compute all three routes and compare. Then work backwards: assume Route C is correct and ρ_{water} is exactly 1.000 g/cm³, and solve for what the displaced volume *should* have been. Compare that against the 6.83 cm³ you measured, express the difference in millilitres, and say whether it is a plausible reading error on a graduated cylinder. Finish by stating which single measurement you would improve first if you ran this again, and how.

AQ2 — the derivation, done properly. Derive $\rho_{water} = \rho_{metal}(1 - T_{water}/T_{air})$ from first principles, stating every assumption as you use it — including the one that is easy to miss, that the cylinder is *fully* submerged so that $V_{displaced} = V_{metal}$. Then explore what happens when that assumption fails: derive the corresponding expression for a partially submerged object, and explain why the volume no longer cancels. What would you have to measure in that case that you did not have to measure here?

AQ3 — propagating the ratio. The scale readings are good to about ±0.005 N. Propagate that uncertainty through the ratio T_{water}/T_{air} and then into ρ_{water} by Route C. (The fractional uncertainty of a quotient is the sum in quadrature of the fractional uncertainties; be careful, because the quantity you need is 1 − the ratio, and its uncertainty is not the same as the ratio's.) Quote Route C's answer with its uncertainty. Then do the same for Route A, propagating ±0.5 mL on the volume. Which route is intrinsically more precise, and by how much?

AQ4 — a cylinder that floats. Suppose you repeated the experiment with a block of wood of density 0.6 g/cm^3, held fully under water by the string. Predict both scale readings qualitatively — and then notice that one of them cannot be what you expected. Work out what the spring scale would actually read, explain the sign of the result physically, and say what it would mean about the direction of the string's force. Does the formula $\rho_{water} = \rho_{metal}(1 - T_{water}/T_{air})$ still hold for this case, and if so what does its output look like?

AQ5 — why a boat floats and a bar of lead does not. Your cylinder lost about 9% of its apparent weight in water. Using your own measured metal density, compute the fraction of its weight any solid object of that density would lose when fully submerged, and show that this fraction is just $\rho_{water}/\rho_{metal}$. Use that to state the condition for floating in a single line. Then resolve the apparent paradox: a steel ship and a steel bar are made of the same material, and one floats. Quantify your answer — estimate the average density a hull must have, and say what fraction of a ship's volume is therefore not steel.

Prompts for your Discussion

These are **not** the Additional Questions. Work them into your **Discussion** section as part of your argument, in your own order. Do not answer them as a list, and do not limit yourself to them.

I. You computed the density of water by three routes. Which do you report as *the* result, and why? A report that gives three numbers without choosing has not finished the analysis.

II. Routes A and B share your displaced-volume measurement. What would it have taken for them to be genuinely independent checks on each other?

III. Your measured metal density differs from the published value for the metal you identified. Is that difference large enough to call the identification into question, or does it point at the measurement instead?

IV. The three trials disagree with one another, particularly in the air reading. Compute the spread and say whether it is large enough to explain your discrepancy from 1.000 g/cm^3.

V. Some water clings to the cylinder when it is lifted out. Which measurement does that affect, in which direction, and is it big enough to matter here?

VI. If the cylinder had touched the bottom of the graduated cylinder during a dip, what would the scale have read and which way would your final answer have moved?

Files for this lab

HANDOUT

This lab, as a PDF

The whole page, printable. The calculator only works on the web version.

REQUIRED

Example workbook

The three trials, a blank analysis sheet laid out for you, and all three routes worked.

GRADING

Lab 08 rubric

The rubric page, with Word and PDF copies. Read it before you write, not after.

SUPPORT

Report starter

Section guidance only — the sentence stems have gone. See the note inside.

NOTE

M6 N1 — Fluids, Pressure and Buoyancy

Where the buoyant force comes from, and Archimedes' principle.

REFERENCE

Formatting data tables

Table captions, units in headers, significant figures.

References to get you started

Your references should include material on buoyancy and Archimedes' principle, and a source for the densities of common metals. Your textbook, this handout and the Module 6 note pages all count, and you need at least one source beyond the textbook.

1. [Author's names], *Journal or Book title*, edition, Volume number, page number, (year).

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